

Discoveries of dual-host affiliated Insect-specific flaviviruses in Zambia

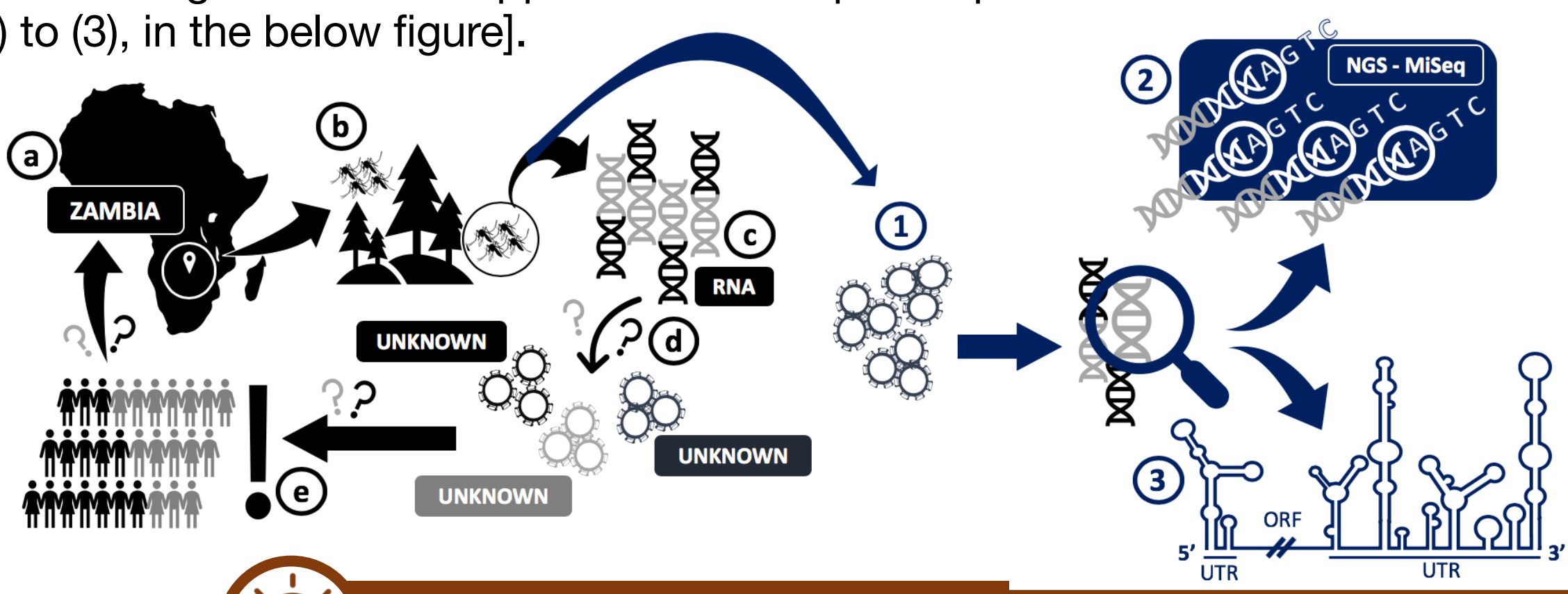
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BACKGROUND

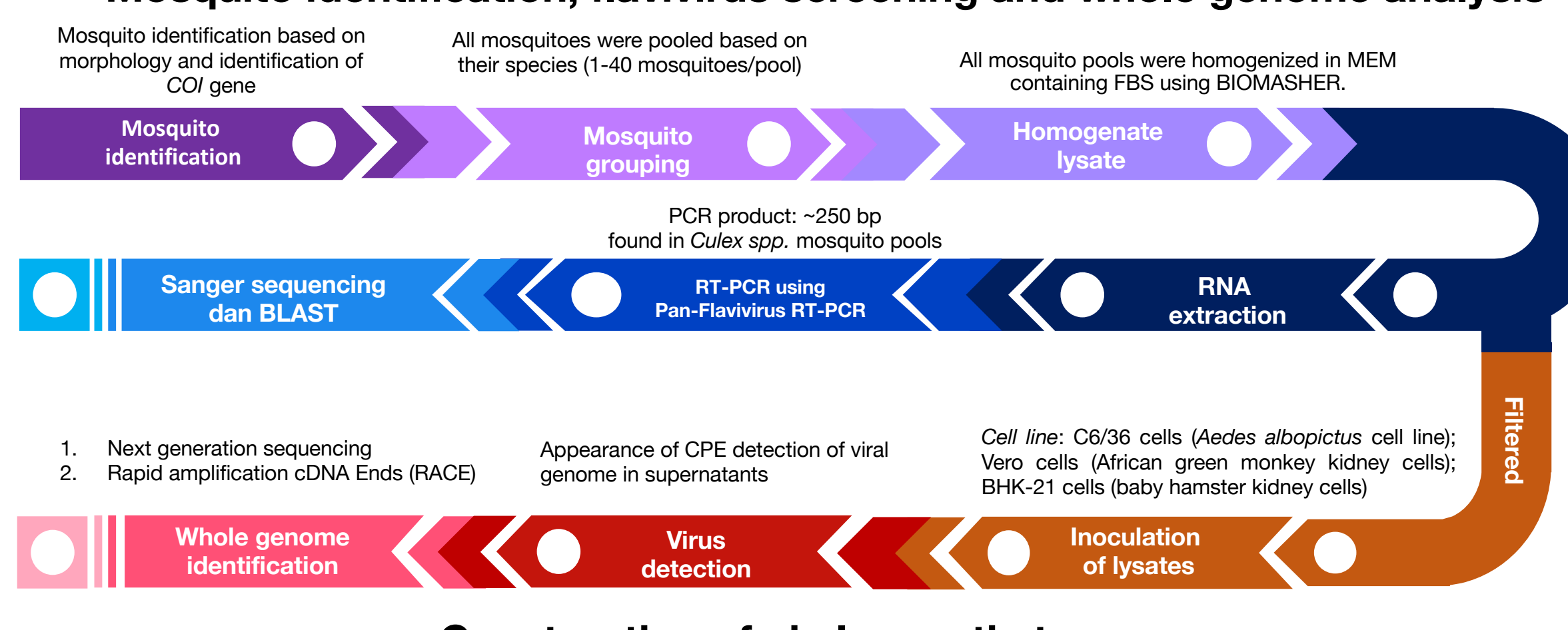
Surveillance of arboviruses in mosquito was conducted in Zambia to provide information for disease control and prevention and to make an informed decision on future disease outbreaks [(a) to (e), in the below figure]. To date, we have collected 15,198 female mosquitoes from various sites in Zambia consisting of several species. We screened for flaviviruses and virus isolation was performed on the positive samples. Whole genome sequencing was performed on the isolated viruses using Illumina Miseq platform and rapid amplification cDNA ends method [(1) to (3), in the below figure].





MATERIALS AND METHODS

Mosquito identification, flavivirus screening and whole genome analysis



Analysis of 5'- and 3'-untranslated regions (UTRs)

We used Covariance Models (CMs) to characterize and find evolutionary conserved elements in viral UTRs. CMs are statistical models that extend Hidden Markov Models (HMMs) by structure information. We built on previously published CMs for dsIFVs (Ochsenreiter et al., 2019) and characterize novel elements based on structural alignments of dsIFV 5'-UTR and 3'UTR regions. *RNAalifold* (from the *ViennaRNA* package) was performed to compute consensus structures of locally stable RNA elements and iteratively build novel CMs that are specific for structured RNAs in dsIFVs UTRs.



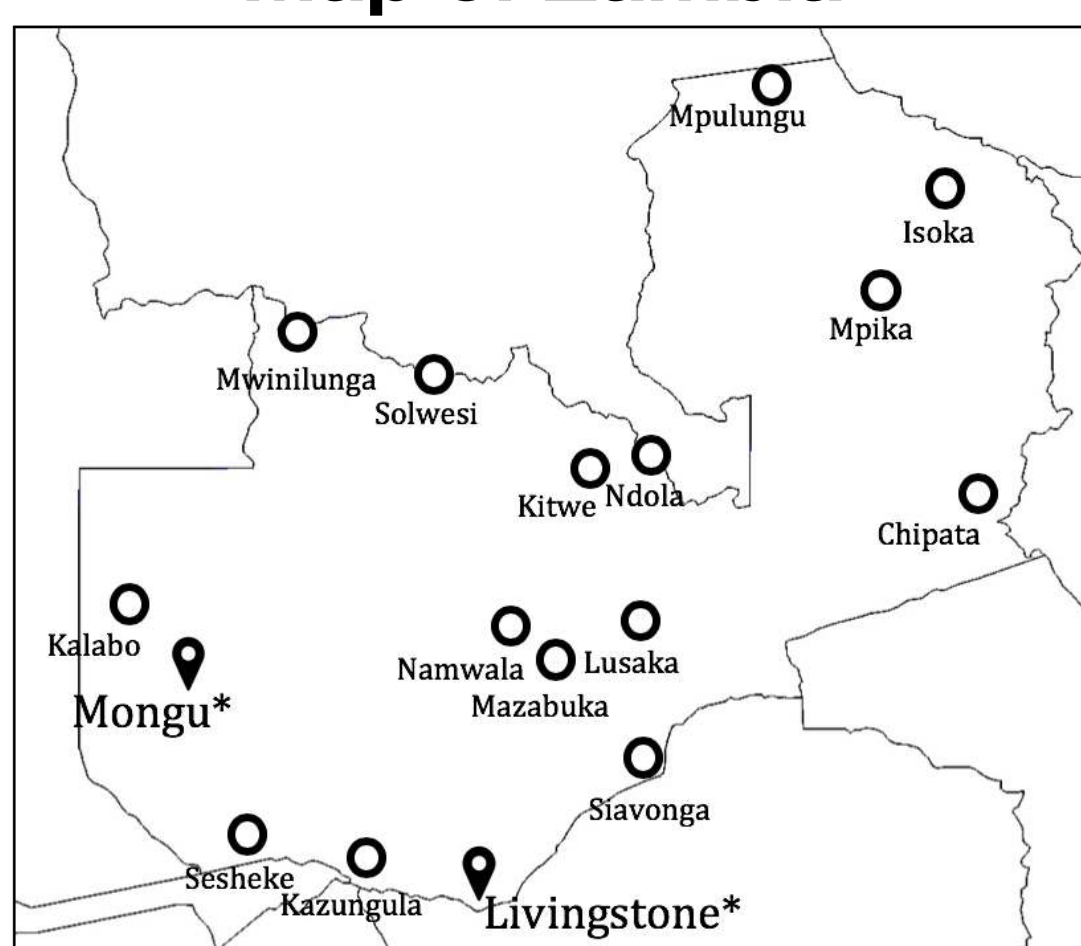
OBJECTIVES

To identify and characterize flaviviruses isolated from mosquitoes collected in several places in Zambia



RESULTS

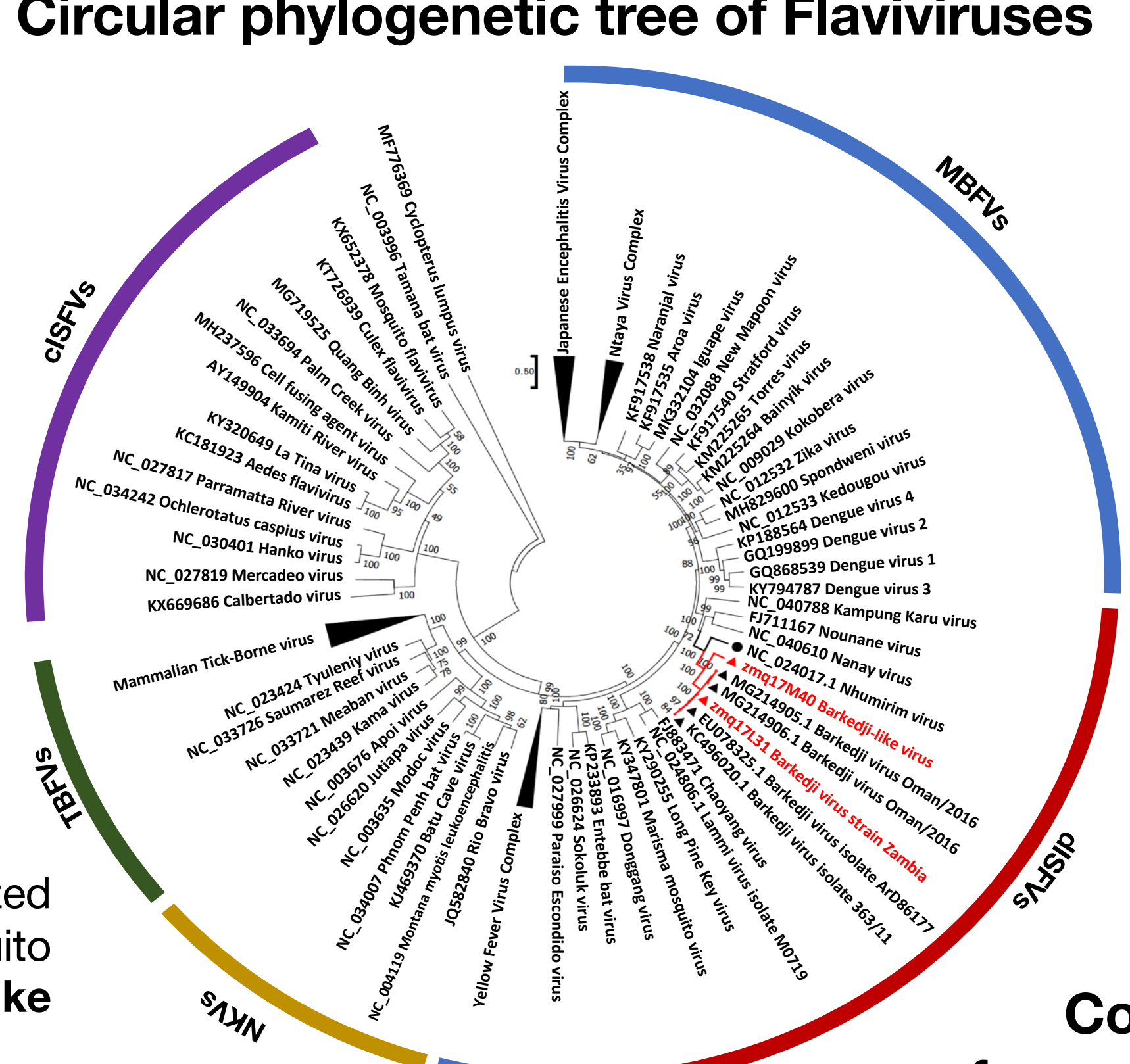
Map of Zambia



Mosquito pools and the prevalence

Year	Month	Place	Species	Mosquito no.	Pool no.	No. positive pools	N5S gene nucleotide identity with BJV Israel			
2017	April	Livingstone	<i>Culex quinquefasciatus</i>	780	34	1 (2.9%)	99%			
			<i>Culex univittatus</i>	15	3					
			<i>Culex nebulosus</i>	5	3					
			<i>Culex tigripes</i>	11	3					
			<i>Culex balenorrhynchus</i>	1	1					
			<i>Aedes aegypti</i>	5	5					
			<i>Aedes spp.</i>	7	3					
			<i>Mansonia sp.</i>	2	2					
			<i>Anopheles spp.</i>	72	15					
			Total	898	69	1				
			2017	May	Mongu	<i>Culex quinquefasciatus</i>	285	22		
						<i>Culex univittatus</i>	309	15	2 (13.3%)	81%
<i>Culex annulirostris</i>	41	5				1 (20%)	81%			
<i>Culex tigripes</i>	2	2								
<i>Culex spp.</i>	109	5								
<i>Aedes triseriatus</i>	10	3								
<i>Aedes aegypti</i>	1	1								
<i>Cosquillettia aurites</i>	9	2								
<i>Cosquillettia metaria</i>	7	2								
<i>Cosquillettia fuscipennis</i>	46	5								
<i>Mansonia sp.</i>	132	8								
<i>Uranotaenia sp.</i>	1	1								
<i>Aedes spp.</i>	10	2								
<i>Anopheles spp.</i>	144	6								
Total	2406	138	3							

Circular phylogenetic tree of Flaviviruses



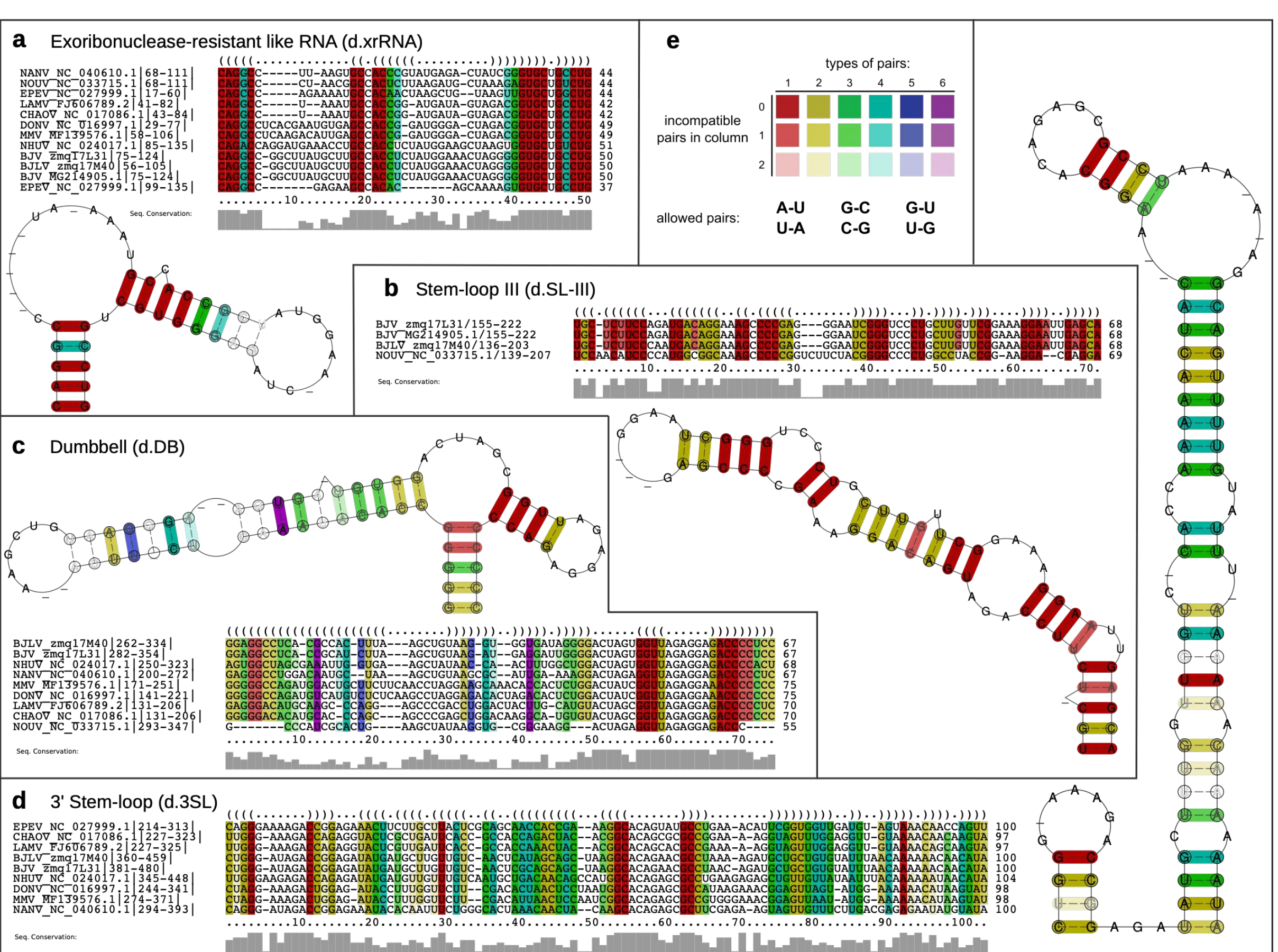
MBFVs: Mosquito-borne flaviviruses
dsIFVs: dual-host insect-specific flaviviruses
cIFSVs: classical insect-specific flaviviruses
NKVs: No-known vector viruses
TBFVs: Tick-borne flaviviruses

Ochsenreiter, Roman, et al. *Viruses* 11.3 (2019): 298.
Liu, Zhong-Yu, et al. *ELIFE* 5 (2016): e17636.
Ziv, Omer, et al. *Nat Methods* 15.10 (2018): 785-788.
Ng, Wy Ching, et al. *Viruses* 9.6 (2017): 137.
Kenney, Joan L., et al. *J Gen Virol* 95.0 12 (2014): 2796.

The importance of 5'- and 3'-UTRs for flavivirus:

- Facilitate the viral genome circularization for replication and translation (Liu, et al., *ELIFE*, 2016).
- Facilitate the virus-host interaction (Ziv O, et al., *Nat Methods*, 2018).
- Host's immune modulation by develop the sub-genomic flavivirus RNA (sfRNA) (Ng WC, et al., *Viruses*, 2017).

Consensus structure prediction of conserved elements in dsIFVs 3'-UTRs



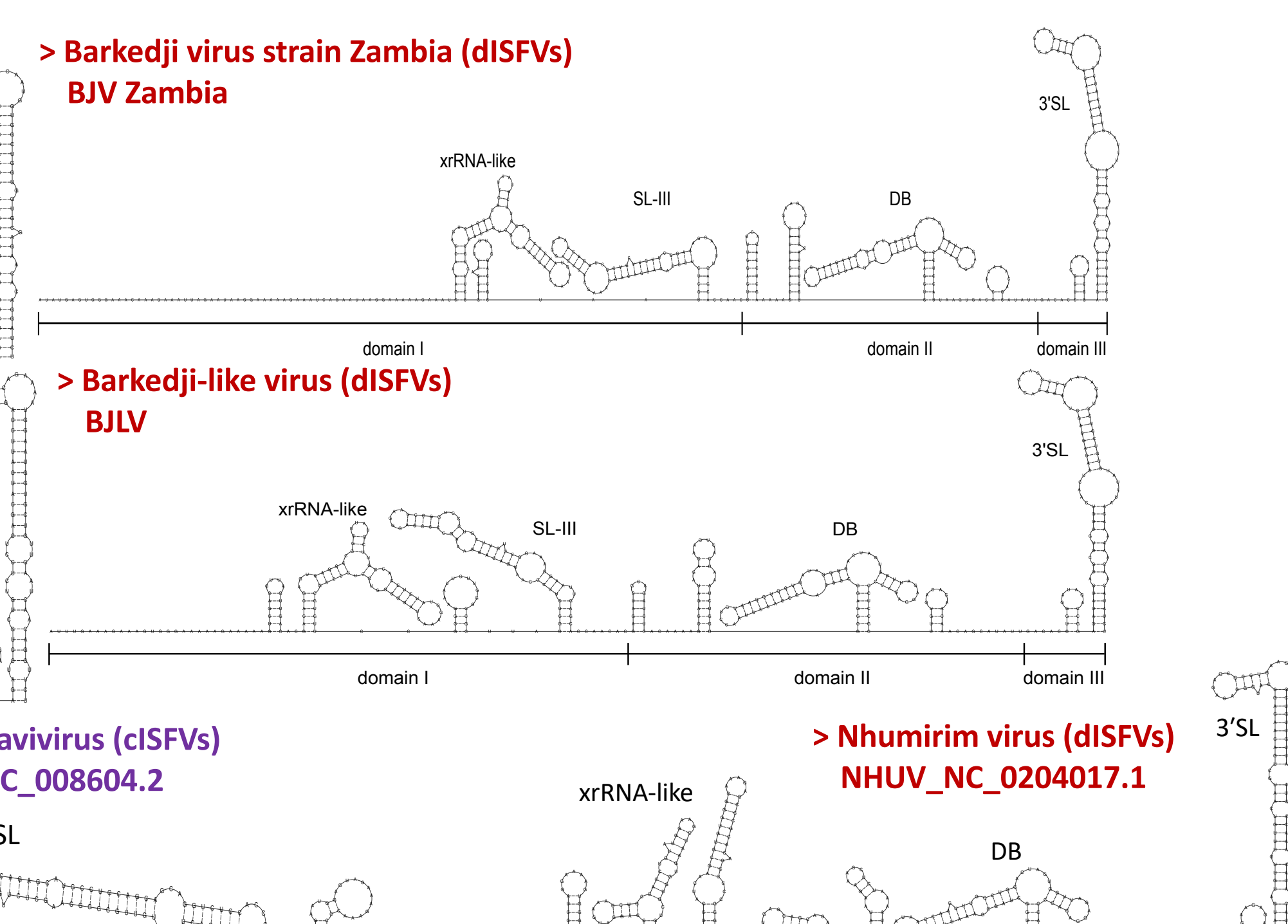
a Exoribonuclease-resistant like RNA (d.xrRNA)
b Stem-loop III (d.SL-III)
c Dumbbell (d.DB)
d 3' Stem-loop (d.3SL)

Consensus structure of conserved RNA structures in dsIFVs 5'-UTRs.

The structural alignments and consensus structure of d.xrRNA (a), d.SL-III (b), d.DB (c) and d.3'SL (e). The gray bars shown below the alignment indicate the sequence conservation. (e) Coloring coding scheme used in *RNAalifold* for paired column in alignment. Circled nucleotides in the consensus structure indicate the compensatory mutations in the corresponding column of the alignment.

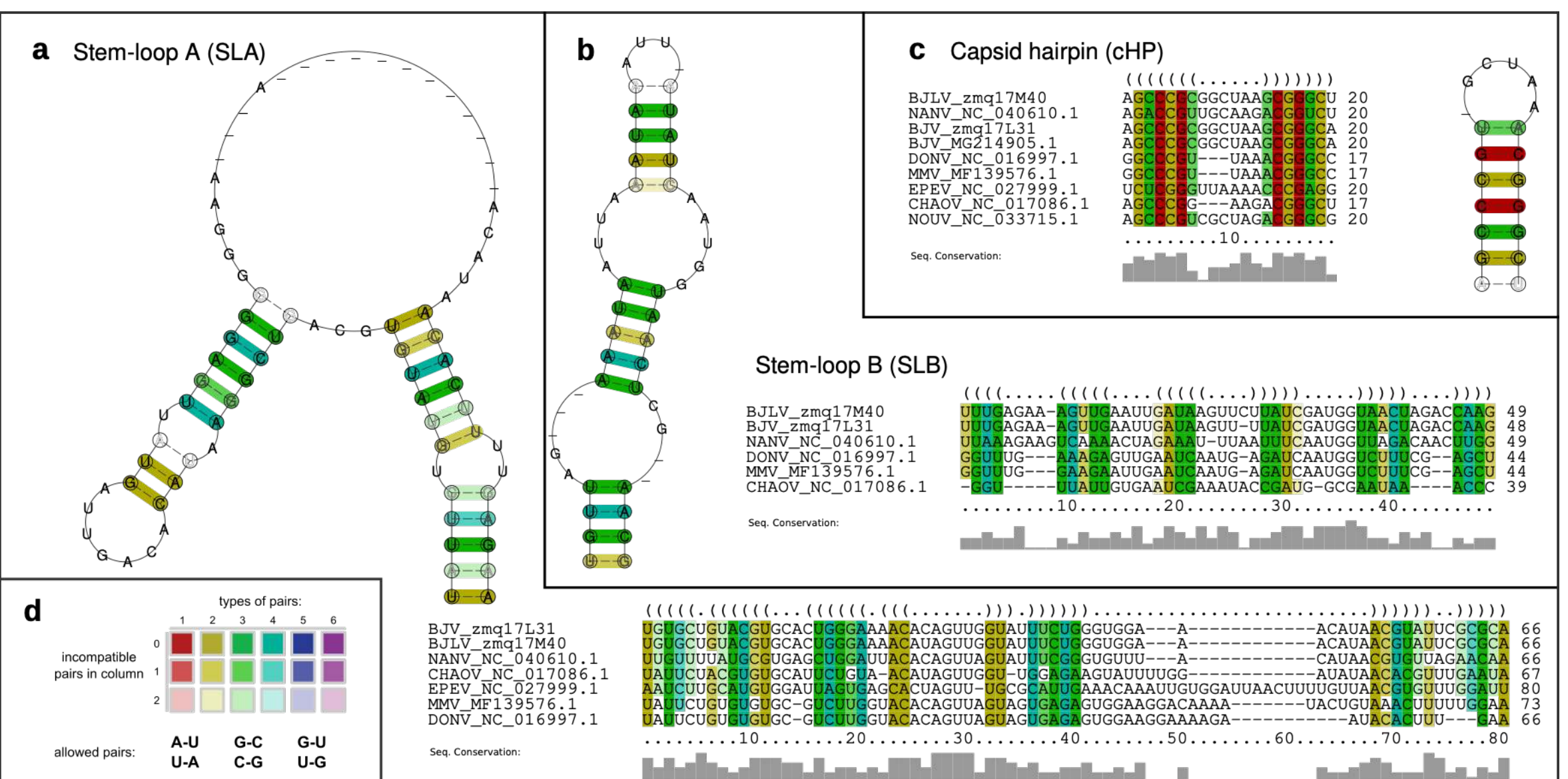
Comparison of 3'-UTRs between dsIFVs (BJV Zambia, BJLV and NHUV), MBFVs (KKUNV and DENV2) and cIFSVs (CxV).

The 3'-UTRs of BJV Zambia and BJLV revealed homologous structures to xrRNA, SL-III, DB and terminal 3'-SL. In comparison, the secondary structure of dsIFVs were more similar to MBFVs rather than cIFSVs.



Consensus structure of conserved RNA structures in dsIFVs 5'-UTRs.

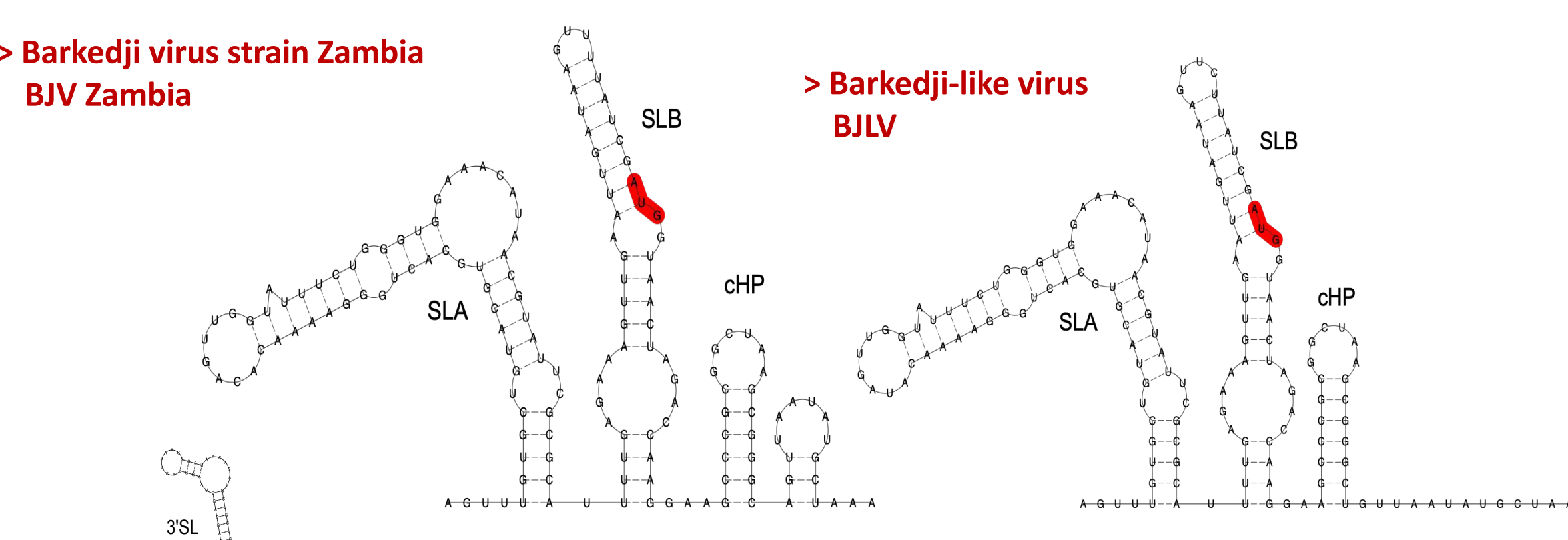
The conserved structure element of dsIFVs 5'-UTRs alongside capsid protein region, SLA (a), SLB (b) and cHP (c). The consensus structure was built from homologous sequence of related viral species. (d) Color coding scheme used in *RNAalifold*; see the 3'-UTRs for details.



a Stem-loop A (SLA)
b Stem-loop B (SLB)
c Capsid hairpin (cHP)

The 5'-UTRs prediction of BJV Zambia and BJLV.

The 5'-UTRs of BJV Zambia and BJLV displayed the canonical structure of 5'-UTRs such as SLA, SLB and, as part of capsid protein nucleotide sequence, cHP. The start codon is highlighted in red.



STRUCTURE ABBREVIATION

SL: stem loop
DB: dumbbell
xrRNA-like: exoribonuclease resistance-like
RCS: repeat conserved-sequence
CS: conserved-sequence
I.Ra/I.Rb: Insect Repeat Element a/b

In this study, we identified and characterized a new strain of **BJV (10,899 nt)** and a novel flavivirus, **BJLV (10,885 nt)** isolated from *Culex* sp. mosquitoes in Zambia. Characterization of 5'- and 3'-UTRs revealed the similarity between dsIFVs and MBFVs through their structure homologs, specifically for BJV Zambia, BJLV and MBFVs. Kenney et al., (*J Gen Virol*, 2014) hypothesized that dsIFVs might be able to replicate in yet to be determined non-insect secondary host. Based on these results, further investigation is needed to **verify the ability of BJV and BJLV to infect and replicate mammalian host.**

COI disclosure: We declare that we have no conflict of interest in relation to this presentation